Ryan R. Hu

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Education

California Institute of Technology (Caltech) — Pasadena, CA

B.S. Computer Science, Minor in Mathematics - *GPA: 4.2/4.0* — *Graduation Date: Jun 2027*

Relevant Coursework: Computer Programming, Programming Methods, Software Design, Deep Learning, Learning Systems, Machine Learning & Data Mining, Algorithms, Computing Systems, Differential Equations, Mathematical Foundations of Computer Science (Discrete Mathematics), Probability and Statistics, Calculus of One and Several Variables and Linear Algebra

Teaching Assistant: Decidability and Tractability (Jan 2025 - Present)

Experience

Machine Learning Intern — Jun 2025 – Present

Tracevision - Math Team — *Los Angeles, CA*

- Developed automated data preprocessing pipelines and scripts to extract and construct shot sequences from raw basketball game footage centered around the basket, optimizing video format and quality for annotation teams through frame-level segmentation and noise reduction.
- Designed and deployed production-grade hybrid RNN + CNN framework for temporal-spatial shot event detection, enabling accurate classification of shot outcomes (made vs. missed) from annotated video clips.

Software Engineer Intern — Apr 2025 – Present

Tracevision - Math Team — *Los Angeles, CA*

- Improved ptools, a full-stack internal tool for object tracking and annotation review, by optimizing frontend and backend components to boost efficiency, reduce latency, and enhance usability. Built with Python, MySQL, HTML/CSS, and JavaScript.
- Automated components of the geospatial API pipeline using a combination of custom scripts and AWS infrastructure (EC2, S3, Lambda), streamlining integration and deployment workflows in security and retail sectors.

Machine Learning Researcher — Feb 2025 – Present

Caltech - Alvarez Lab — *Pasadena, CA*

- Developed a custom autoencoder architecture to detect anomalous player behavior by analyzing reconstruction error as a proxy for behavioral deviation from typical gameplay patterns in Activision's Call of Duty (COD) games.
- Utilized SQL queries and implemented Python scripts to clean raw gameplay logs and extract additional input features, expanding autoencoder representation space and improving anomaly sensitivity.
- Developed a verification pipeline combining clustering techniques with time-based segmentation to evaluate autoencoder performance and assess how different types of anomalous behavior affect player engagement and retention.

Undergraduate Researcher — Apr 2024 – Dec 2024

Caltech - Alvarez Lab — *Pasadena, CA*

- Leveraged causal machine learning models and gradient-boosting trees to analyze the role of socioeconomic and epigenetic factors in the aging of people living with HIV from patient data collected by the Multicenter AIDS Cohort Study (MACS).
- Built data preprocessing pipeline for cleaning and transforming inputs for efficiency in machine learning tasks downstream.
- Research conducted as part of Caltech's Summer Undergraduate Research Fellowship (SURF) program in collaboration with UCLA Epigenetics. Currently authoring a manuscript based on findings.

Undergraduate Researcher — Jun 2023 – Sep 2023

Caltech - GALCIT Lab — *Pasadena, CA*

- Developed a visual anemometry framework in MATLAB combining YOLOv3 object detection and optical flow mapping techniques to estimate wind speed and direction from canopy motion, enabling contact-free environmental sensing.
- Conducted a quantitative analysis linking vegetation displacement variance to wind standard deviation, achieving explainable correlation patterns across different tree species and wind conditions.

Selected Projects

Agentic Stock Trader | *Python*

- Built a multi-agent stock trading framework where LLM-driven agents ingest financial news, debate with each other, and output explainable decisions.
- Designed a recursive agent debate architecture with static personas, dynamic leaf agents, and hierarchical clustering.
- Developed an evaluation pipeline to calculate execution slippage, enabling benchmarking in messy, highly volatile environments.

Endless Runner Video Game | *C, HTML/CSS, SDL*

- Developed Google Dino Run-inspired endless runner game with obstacles, powerups, and progressive difficulty scaling. Written entirely in C using standard libraries.
- Implemented unit tests for physics engine. Reduced collision computation overhead by restricting hitbox checks only active screen segments with runtime bounds culling.

Activities/Achievements

NCAA Division III Men's Basketball — Sep 2023 – April 2025

SCIAC All-Academic Team — 2024, 2025

Technical Skills

Languages: Python, C, C++, Java, JavaScript/TypeScript, HTML/CSS, MATLAB, LaTeX

Developer Tools: Git, VS Code, AWS, Github, Linux, Docker, Kubernetes, Figma

Frameworks/Libraries: React.js, Node.js, Pandas, OpenCV, PyTorch, Keras, TensorFlow, Scikit, Tailwind, Flask, Plotly

Skills: CI/CD, Machine Learning, Full Stack, Software Engineering, Data Analysis